

```
!pip install spacy pandas matplotlib
!python -m spacy download en_core_web_sm
```

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Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packa
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Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packag
Collecting en-core-web-sm==3.8.0
```

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0.tar.gz 12.8/12.8 MB 98.0 MB/s eta 0:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the

```
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
from spacy.matcher import Matcher
```

```
nlp = spacy.load("en_core_web_sm")
```

```
# Example: CSV containing a column named 'abstract'
data = pd.read_csv('arxiv_data.csv', on_bad_lines='skip', engine='python')

# Select AI-related abstracts (sample)
abstracts = data['titles'].dropna().head(50)
```

```
docs = [nlp(text) for text in abstracts]
```

```
sample_doc = docs[0]
tokens = [token.text for token in sample_doc]

print(tokens[:30])
```

```
['Survey', 'on', 'Semantic', 'Stereo', 'Matching', '/', 'Semantic', 'Depth',
```

```
noun_phrases = []

for doc in docs:
    for chunk in doc.noun_chunks:
        noun_phrases.append(chunk.text.lower())

np_freq = Counter(noun_phrases)
top_noun_phrases = np_freq.most_common(10)

print("Top Noun Phrases:")
for np, count in top_noun_phrases:
    print(np, ":", count)
```

```
Top Noun Phrases:
semi-supervised medical image segmentation : 4
medical image segmentation : 3
semantic segmentation : 3
transformer : 2
image segmentation : 2
semantic stereo matching / semantic depth estimation : 1
future-ai: guiding principles : 1
consensus recommendations : 1
trustworthy artificial intelligence : 1
future medical imaging : 1
```

```

entities = []

for doc in docs:
    for ent in doc.ents:
        if ent.label_ in ["ORG", "DATE", "PRODUCT", "GPE"]:
            entities.append(ent.text)

entity_freq = Counter(entities)
print(entity_freq.most_common(10))

```

```
[('FUTURE-AI: Guiding Principles and Consensus Recommendations for Trustworth
```

```

matcher = Matcher(nlp.vocab)

# Pattern: Adjective + Noun (e.g., "deep learning")
pattern1 = [{"POS": "ADJ"}, {"POS": "NOUN"}]

# Pattern: Noun + Noun (e.g., "neural network")
pattern2 = [{"POS": "NOUN"}, {"POS": "NOUN"}]

matcher.add("TECH_TERMS", [pattern1, pattern2])

matches = []
for doc in docs:
    for match_id, start, end in matcher(doc):
        matches.append(doc[start:end].text)

print("Sample Technical Terms:")
print(matches[:10])

```

```

Sample Technical Terms:
['supervised 3D', 'operational context', 'medical image', 'image segmentation

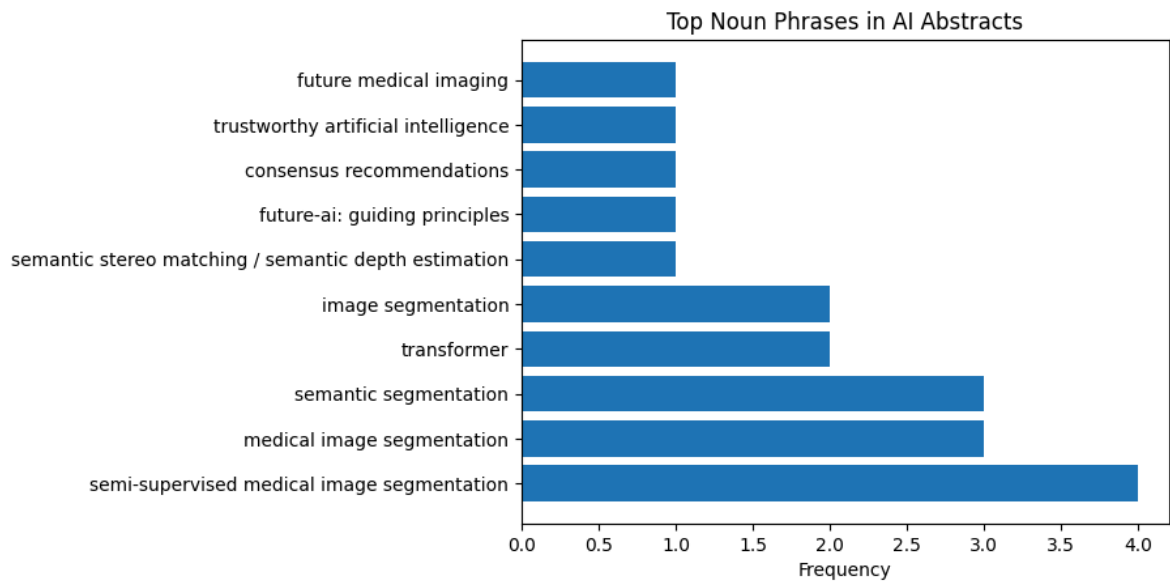
```

```

phrases, counts = zip(*top_noun_phrases)

plt.figure()
plt.barh(phrases, counts)
plt.xlabel("Frequency")
plt.title("Top Noun Phrases in AI Abstracts")
plt.show()

```



```
entity_items = entity_freq.most_common(10)
entities, ent_counts = zip(*entity_items)

plt.figure()
plt.bar(entities, ent_counts)
plt.xticks(rotation=45)
plt.title("Named Entity Frequency")
plt.show()
```

